Advanced Econometrics Lecture 03

dr Marcin Chlebus, dr Rafał Woźniak

Faculty of Economic Sciences, University of Warsaw

Warsaw

Today's agenda

1. Quick intro
   - All you have to know to pass this course
   - Logit and probit
   - Marginal effects
   - Linktest and other diagnostic tests
   - Hypotheses verification – LR test
   - Linear Probability Model
2. Details
   - Nice to know
   - The stuff we usually teach our students majoring in econometrics
   - Details of the models
   - Derivation of marginal effects
   - Maximum Likelihood Method
   - How these models are estimated
   - Information Criteria
   - Cut-off probabilities

Binary Dependent Variables - [1] p. 418

- [In economics] we focus primarily on the dependent variables which are continuous and fully observable; quantities, prices, and outputs are examples such variables.
- Microeconomics is a general theory of choice, and many of the choices that individuals and firms make cannot be measured by a continuous outcome variable.
- Something/Someone fulfils/ does not fulfil some condition.
- Let's assume that we are interested in factors that impact on a variable describing that a student failed at least one exam during studies or not.
- Might such variable depend on *year of college*?

Decoding. Binary Dependent Variables

- y id defined in the way that something/someone fulfils/does not fulfil some condition.
- According to the binary choice models convention while decoding dependent variable 0 is failure and 1 is success.
- **Question:** What is a failure and a success in the example of students failing exams?
- How can we decode income, height, price into binary dependent variables?
- Does it make sense? Always?

Qualitative, Limited or Count Dependent Variable

1. Models with Binary Dependent Variable
   - Logit
   - Probit
   - Linear Probability Model
2. Unoredered Choice Models
   - Multinomial Logit
   - Conditional Logit
3. Ordered Choice Models
   - Ordered Logit
   - Ordered Probit
4. Models for Count Data
   - Poisson Regression Models
   - Negative Binomial Models

Models with Binary Dependent Variable - [2] p. 772

- To focus ideas, consider the model of labour force participation [...].
- The respondent either works or seeks work ($Y = 1$) or does not ($Y = 0$) in the period in which our survey is taken.
- We believe that a set of factors, such as age, marital status, education, and work history, gathered in a vector $\boldsymbol{x}$ explain the decision, so that

$$Prob(Y = 1|\mathbf{x}) = F(\boldsymbol{x}, \boldsymbol{\beta}) \quad (1)$$

$$Prob(Y = 0|\mathbf{x}) = 1 - F(\boldsymbol{x}, \boldsymbol{\beta}) \quad (2)$$

- The set of parameters $\boldsymbol{\beta}$ reflects the impact of changes in $\boldsymbol{x}$ on the probability.
- For example, among the factors that might interest us is the marginal effect of marital status on the probability of labour force participation.
- The problem at this point is to devise a suitable model for the right-hand side of the equation.

Models with Binary Dependent Variable

In binary choice models we try to explain probability of success with explanatory variables.

$$p = Prob(Y = 1|\mathbf{x}) = F(\boldsymbol{x}, \boldsymbol{\beta}) \tag{3}$$

What is the form of $p(x_i) = F(\boldsymbol{x}, \boldsymbol{\beta})$? This should be a function that takes values in the 0-1 interval.

$p(x_i)$ is more often than not a cumulative distribution function of some continuous distribution.

Models with Binary Dependent Variable

- **Probit**
  - In the probit model we assume that $F(.)$ is cumulative distribution function of the standard normal distribution.
  - The probability of success is equal to $\Phi(x_i\beta)$, where $\Phi(.)$ cumulative distribution function of $N(0,1)$.
- **Logit**
  - In the logit model we assume that $F(.)$ is cumulative distribution function of the logistic distribution.
  - The probability of success is equal to $\Lambda(x_i\beta)$.

Example - oscar.csv - dataset from [1]

variable name	variable label
year	year
title	movie title
winner	= 1 film won Oscar for best picture
nominations	= total number of Oscar nominations
gglobes	= number of Golden Globe Awards won
comedy	=1 if comedy

Logit model

```
Call:
glm(formula = winner ~ nominations + gglobes, family = binomial(link = "logit"),
    data = dataset)

Deviance Residuals:
     Min        1Q    Median        3Q       Max
-1.81358  -0.27738  -0.13811  -0.05024   2.66524

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -9.3768     2.0533  -4.567 4.95e-06 ***
nominations   0.6756     0.1914   3.530 0.000415 ***
gglobes       1.1252     0.3265   3.446 0.000568 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 100.080  on 99  degrees of freedom
Residual deviance:  45.782  on 97  degrees of freedom
AIC: 51.782

Number of Fisher Scoring iterations: 6
```

Marginal effects

```
Call:
logitmfx(formula = winner ~ nominations + gglobes, data = dataset,
          atmean=TRUE)

Marginal Effects:
               dF/dx Std. Err.       z    P>|z|
nominations 0.033897  0.015135 2.2397 0.02511 *
gglobes     0.056457  0.025163 2.2436 0.02486 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

The code and the output above presents marginal effects calculated for means of variables *nominations* and *gglobes*.

Unfortunately, there is no package in R that allows user to calculate marginal effects for user selected values of x.

marginaleffects.R

```
model <- glm(winner~nominations+gglobes, data = oscar,
family = binomial(link = "logit"))
summary(model)

x = c(1,7.43,1.31) #convention: (intercept, x1, x2, ...)
marginaleffects(model, x)
```

```
> x = c(1,7.43,1.31) #convention: (intercept, x1, x2, ...)
> marginaleffects(model, x)
            Marginal effects at X=
nominations       0.03389660  7.43
gglobes           0.05645656  1.31
```

Measuring goodness of fit - [2] p.790

There have been many fit measures suggested for qualitative response (QR) models. An analog to the R^2 in a conventional regression is McFadden's likelihood ratio index,

$$LRI = 1 - \frac{\ln L}{\ln L_0}. \tag{4}$$

This measure has an intuitive appeal in that it is bounded by zero and one. If all slope coefficients are zero, then it equals zero.

Unfortunately, the value between zero and one have no natural interpretation.

```
PseudoR2(mylogit)
       McFadden     Adj.McFadden        Cox.Snell
      0.5425491        0.4626134        0.4189891
     Nagelkerke McKelvey.Zavoina           Effron            Count
      0.6625208        0.7006450        0.5530570        0.9000000
      Adj.Count              AIC    Corrected.AIC
      0.5000000       51.7819082       52.0319082
```

LINKTEST

1. LINKTEST test
    - $H_0 :$ $\qquad y_i = f\left(x_i\beta\right) + \varepsilon_i,$
    - $H_0 :$ $\qquad y_i = g\left(x_i\beta\right) + \varepsilon_i,$
    - where $g(.) \neq f(.)$.
2. Step 1
    - Regression $y_i = f\left(x_i\beta\right) + \varepsilon_i,$
    - Fitted values $\hat{y}_i$.
3. Step 2
    - Auxiliary regression $y_i = \alpha_0 + \alpha_1\hat{y}_i + \alpha_2\hat{y}_i^2 + u_i,$
    - Test $\alpha_2 = 0$.

linktest.R

```
source("linktest.R")

linktest_result = linktest(model)
summary(linktest_result)

> summary(linktest_result)
Call:
glm(formula = y ~ yhat + yhat2,
family = binomial(link = model$family$link))
Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.03406    0.44867  -0.076 0.939490
yhat         1.03227    0.28856   3.577 0.000347 ***
yhat2        0.01598    0.08737   0.183 0.854863
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Diagnostic tests of logits and probits

- **Model's specification is correct**
- The linktest
  - `source("linktest.R")`
- The Hosmer-Lemeshow test
  - `generalhoslem` package
  - `https://www.chrisbilder.com/categorical/Chapter5/AllGOFTests.R`
- The Osius-Rojek test
  - `https://www.chrisbilder.com/categorical/Chapter5/AllGOFTests.R`
- The Stukel test
  - `https://www.chrisbilder.com/categorical/Chapter5/AllGOFTests.R`
- Example: `Diagnostic_tests_for_DDV.R`

```
library("wooldridge")
library("generalhoslem")
data(mroz, package="wooldridge")

# a logit model
logit = glm(inlf~nwifeinc+educ+exper+I(exper^2)+
            age+kidslt6+kidsge6,
            data=mroz, family=binomial(link="logit"))
summary(logit)

# the Hosmer-Lemeshow test
logitgof(mroz$inlf, fitted(logit), g = 10)

# load functions from AllGOFTests.R
# the Osius-Rojek test
o.r.test(logit)

# the Stukel test
stukel.test(logit)
```

```
> logitgof(mroz$inlf, fitted(logit), g = 10)

Hosmer and Lemeshow test (binary model)

data:  mroz$inlf, fitted(logit)
X-squared = 12.851, df = 8, p-value = 0.1171

> o.r.test(logit)
z =  -0.5722773 with p-value =  0.5671342

> stukel.test(logit)
Stukel Test Stat =  3.66528 with p-value =  0.1599907
```

Hypotheses testing

Three general procedures used in developing statistics to test hypotheses

- the likelihood ratio test (**L**ikelihood **R**atio)
- the Wald test
- the Lagrange multiplier test LM (**L**agrange **M**ultipliers)

Tested hypothesis

$$H_0 : \begin{cases} h_1(\theta) = 0 \\ \vdots \\ h_g(\theta) = 0 \end{cases} \tag{5}$$

where g is the number of restrictions.

Likelihood Ratio Test - [2]

- Notation
  - $\widehat{\theta}$ - unrestricted maximum likelihood estimator
  - $\widehat{\theta}_R$ - restricted maximum likelihood estimator
- $LR = 2\left[\ln L(\widehat{\theta}) - \ln L(\widehat{\theta}_R)\right] \to \chi^2_g$
- Easy to calculate.
- For larger values of LR the restrictions under the null hypothesis are not consistent with the data since a serious loss of information caused by imposing these restrictions now results. Then, the null is rejected in favour of the alternative hypothesis.
- The LR test requires estimating the model under both the null and alternative hypotheses to obtain $l(\widehat{\theta})$ and $l(\widehat{\theta}_R)$.

Likelihood Ratio Test – Example

```
myprobit <- glm(winner~nominations+gglobes, data=oscar,
                family=binomial(link="probit"))
summary(myprobit)

myprobit_restricted <- glm(winner~nominations, data=oscar,
                           family=binomial(link="probit"))
summary(myprobit_restricted)

lrtest(myprobit, myprobit_restricted)
> lrtest(myprobit, myprobit_restricted)
Likelihood ratio test

Model 1: winner ~ nominations + gglobes
Model 2: winner ~ nominations
  #Df LogLik Df  Chisq Pr(>Chisq)
1   3 -22.75
2   2 -30.35 -1 15.199  9.676e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Linear Probability Model - [2] p. 772

One possibility [to model binary dependent variable] is to retain the familiar linear regression,

$$F(\boldsymbol{x}, \boldsymbol{\beta}) = \boldsymbol{x}'\boldsymbol{\beta}. \tag{6}$$

Because $\mathbb{E}(y|\boldsymbol{x}) = F(\boldsymbol{x}, \boldsymbol{\beta})$, we can construct the regression model ,

$$y = \mathbb{E}(y|\boldsymbol{x}) + (y - \mathbb{E}(y|\boldsymbol{x})) = \boldsymbol{x}'\boldsymbol{\beta} + \varepsilon. \tag{7}$$

The **linear probability model** has a number of shortcomings.

- ε is heteroscedastic
- We cannot constraint $\boldsymbol{x}'\boldsymbol{\beta}$ to the 0-1 interval.

Please, do not use the linear probability model.

Linear Probability Model – just don't!

Quick intro

1. The end of this quick intro
    - All you have to know to pass this course
    - Logit and probit
    - Marginal effects
    - Linktest and other diagnostic tests
    - Hypotheses verification – LR test
    - Linear Probability Model

To sum up

1. Models
   - Logit, probit, linear probability model (LPM)
2. Interpretation
   - Parameters: Only signs
   - Marginal effects for:
     - any characteristics
     - average marginal effects (AME)
     - marginal effects for means (MEM)
   - R^2 statistics
3. Tests
   - Linktest, Hosmer-Lemeshow, Stukel, Osius-Rojek, Likelihood ratio test
4. Model selection
   - Specification tests, Significance of parameters, Information Criteria
5. Cut-off probabilities

Nice if you knew, that is, for the highest grade

2. Details
   - Nice to know
   - The stuff we usually teach our students majoring in econometrics
   - Details of the models
   - Derivation of marginal effects
   - Maximum Likelihood Method
   - How these models are estimated
   - Information Criteria
   - Cut-off probabilities

How does it work? - [1] p. 419

The probability function for for a binary random variable is

$$f(y) = p^y (1-p)^{1-y}, \quad y = 0, 1 \tag{8}$$

where p is the probability that y takes the value 1. This discrete random variable has expected value $\mathbb{E}(y) = p$ and variance $var(y) = p(1-p)$.

$$p = Prob(Y = 1|\mathbf{x}) = F(\boldsymbol{x}, \boldsymbol{\beta}) \tag{9}$$

The set of parameters $\boldsymbol{\beta}$ reflects the impact of changes in $\boldsymbol{x}$ on the probability.

How does it work? - [1] p. 419

In binary choice models we model *the latent variable*

$$
\begin{aligned}
y^* &= \boldsymbol{X}\beta + \varepsilon && (10)\\
\hat{p} &= \hat{y}^* = F(\boldsymbol{x}, \hat{\boldsymbol{\beta}}). && (11)
\end{aligned}
$$

On the basis of the values of the p^* we generate predicted values

$$
\begin{aligned}
\hat{y} &= 1 \quad \text{if} \quad \hat{p} \geq p^* && (12)\\
\hat{y} &= 0 \quad \text{if} \quad \hat{p} < p^* && (13)
\end{aligned}
$$

The set of parameters $\boldsymbol{\beta}$ reflects the impact of changes in $\boldsymbol{x}$ on the probability.

Estimation

The likelihood function of the binary choice model for cross-section data

$$L(\theta|\boldsymbol{x}) = \prod_{i=1}^{N} [p(x_i)]^{y_i} [1 - p(x_i)]^{1-y_i} \tag{14}$$

Its logarithm has the form

$$\ln(\theta|\boldsymbol{x}) = \ln L(\theta|\boldsymbol{x}) = \sum_{i=1}^{N} y_i \ln p(x_i) + \sum_{i=1}^{N} (1 - y_i) \ln[1 - p(x_i)] \tag{15}$$

Since $p(x_i) \in [0, 1]$, it is easy to show that $\ln(\theta|\boldsymbol{x}) \leq 0$.

Find $\hat{\theta}$ using maximum likelihood method.

Maximum Likelihood Method

Let $f(x;\theta)$ denote the pdf of a continuous random variable with associated parameter θ.
Suppose $X_1, X_2, \ldots, X_n$ are a random sample from this distribution and $x_1, x_2, \ldots, x_n$ are actual observed values.

Then, the *likelihood function* of θ is

$$L(\theta|x_1, x_2, \ldots, x_n) = f(x_1;\theta)f(x_2;\theta)\cdots f(x_n;\theta)$$

A maximum likelihood estimate is a statistic $\hat{\theta}$ that maximises $L(\theta)$

$$L(\hat{\theta}) \geq L(\theta) \text{ for all } \theta. \tag{16}$$

Source: *Mathematical Statistics with Resampling and R, str. 139.*

Likelihood function for the Oscar probit model

Let's assume a simple model for Oscar movies

$$oscar_i = \beta_0 + \beta_1 nominations_i + \varepsilon_i \tag{17}$$

Probability of the success is then

$$Prob(oscar_i = y_i) = [\Phi(\beta_0 + \beta_1 nom_i)]^{y_i} [1 - \Phi(\beta_0 + \beta_1 nom_i)]^{1-y_i} \tag{18}$$

Likelihood function is then

$$L(\beta_0, \beta_1) = \prod_{i=1}^{N} [\Phi(\beta_0 + \beta_1 nom_i)]^{y_i} [1 - \Phi(\beta_0 + \beta_1 nom_i)]^{1-y_i} \tag{19}$$

And the log-likelihood function

$$lnL(\beta_0, \beta_1) = \sum_{i=1}^{N} y_i \Phi(\beta_0 + \beta_1 nom_i) + \sum_{i=1}^{N} (1 - y_i) [1 - \Phi(\beta_0 + \beta_1 nom_i)] \tag{20}$$

Likelihood function for the Oscar probit model

Likelihood function is then

$$L(\beta_0, \beta_1) = \prod_{i=1}^{N} \left[\Phi(\beta_0 + \beta_1 nom_i)\right]^{y_i} \left[1 - \Phi(\beta_0 + \beta_1 nom_i)\right]^{1-y_i} \tag{21}$$

And the log-likelihood function

$$lnL(\beta_0, \beta_1) = \sum_{i=1}^{N} y_i \Phi(\beta_0 + \beta_1 nom_i) + \sum_{i=1}^{N} (1 - y_i) \left[1 - \Phi(\beta_0 + \beta_1 nom_i)\right] \tag{22}$$

Now, we have to maximise this function with regard to β_0 and β_1.

More on the topic

- To extend the material discussed I highly recommend these:
- https://statisticalhorizons.com/hosmer-lemeshow
- https://statisticalhorizons.com/alternatives-to-the-hosmer-lemeshow-test
- https://davegiles.blogspot.com/2012/06/another-gripe-about-linear-probability.html
- https://davegiles.blogspot.com/2011/05/gripe-of-day.html
- https://davegiles.blogspot.com/2013/05/robust-standard-errors-for-nonlinear.html#more

How to interpret?

Index function models
We assume that there is an latent index that impacts the probability of success. This interpretation is used if observed y_i cannot be interpreted as decisions of entities and x contains only entities' characteristics.

Random utility models
Successes and failures are results of observed entities' decisions. If consumer choices are regarded then their utility is maximised. In case of firms, profit is maximised. We try to reconstruct the way the decisions are taken.

Inference in binary choice models

Binary choice models are not linear.

$$\mathbb{E}(y|x) = 0[1 - F(x\beta)] + 1F(x\beta) = F(x\beta) = p(x\beta) \tag{23}$$

Marginal effect

$$\frac{\partial \mathbb{E}(y|x)}{\partial x_k} = \frac{\partial p(x\beta)}{\partial x_k} = f(x\beta)\beta_k \tag{24}$$

Marginal effects are interpreted as the expected change of the probability of success due to unitary change in the explanatory variable.

Inference in binary choice models

$$\frac{\partial \mathbb{E}(y|x)}{\partial x_k} = \frac{\partial p(x\beta)}{\partial x_k} = f(x\beta)\beta_k \tag{25}$$

- Marginal effect depends on x.
- Usually, marginal effects are calculated and presented for mean of x.
- Marginal effect for a dummy variable is defined as $\Delta F(\bar{x}\beta) = F(\bar{x}_1\beta) - F(\bar{x}_0\beta)$.
- Elements of the vector β do not have interpretation.
- Ratio between two parameters is interpretable

$$\frac{\partial E(y|x)}{\partial x_k} / \frac{\partial E(y|x)}{\partial x_m} = \frac{\beta_k}{\beta_m} \tag{26}$$

- and independent of $\boldsymbol{x}$.

Classification table

		Observed		
		0	1	Total
Predicted	0	n_{00}	n_{01}	$n_{00}+n_{01}$
	1	n_{10}	n_{11}	$n_{10}+n_{11}$
	Total	$n_{00}+n_{10}$	$n_{01}+n_{11}$	n

Sensitivity

$$Pr\left(\hat{p}_i \geq p^* | y_i = 1\right) \approx \frac{n_{11}}{n_{01} + n_{11}} \tag{27}$$

Specificity

$$Pr\left(\hat{p}_i < p^* | y_i = 0\right) \approx \frac{n_{00}}{n_{00} + n_{10}} \tag{28}$$

Measuring goodness of fit

McKelvey & Zavoina

$$R^2_{MZ} = \frac{\sum_{i=1}^{n}\left(\boldsymbol{x}_i'\hat{\boldsymbol{\beta}} - \bar{\boldsymbol{x}}_i'\hat{\boldsymbol{\beta}}\right)^2}{n + \sum_{i=1}^{n}\left(\boldsymbol{x}_i'\hat{\boldsymbol{\beta}} - \bar{\boldsymbol{x}}_i'\hat{\boldsymbol{\beta}}\right)^2} \tag{29}$$

Count R^2

$$R^2_{count} = \frac{\#correct}{n} \tag{30}$$

Adjusted count R^2

$$R^2_{Adj.-count} = \frac{\#correct - m}{n - m} \tag{31}$$

where m is the most frequent outcome from the classification table.

Source: https://stats.idre.ucla.edu/ (link)

Odds Ratio

$$Odds(x_i) = \frac{\mathbb{P}(y_i = 1|x_i)}{\mathbb{P}(y_i = 0|x_i)} \tag{32}$$

If a student have not studied the whole textbook, then the probability of passing the exam is $\frac{1}{9}$, while studied is $\frac{8}{9}$. Thus odds of passing the exam is $Odds(0) = \frac{1/9}{8/9} = \frac{1}{8}$.

Odds ratio for success for i-th and j-th observations is equal to

$$\begin{aligned} \frac{Odds(x_i)}{Odds(x_j)} &= \exp\left[(x_i - x_j)\beta\right] = \exp\left(\Delta x \beta\right) \\ &= \exp\left(\Delta x_1 \beta_1 + \ldots + \Delta x_K \beta_K\right) \end{aligned} \tag{33}$$

Log Odds

In the logit model the link function is the logistic cumulative distribution function.

$$F(\boldsymbol{x}, \boldsymbol{\beta}) = \Lambda(\boldsymbol{x}, \boldsymbol{\beta}) = \frac{\exp(\boldsymbol{x}'\boldsymbol{\beta})}{1 + \exp(\boldsymbol{x}'\boldsymbol{\beta})} \tag{34}$$

Assume model

$$y^* = \beta_0 + \beta_1 x_1 + \ldots + \beta_K x_K + \varepsilon \tag{35}$$

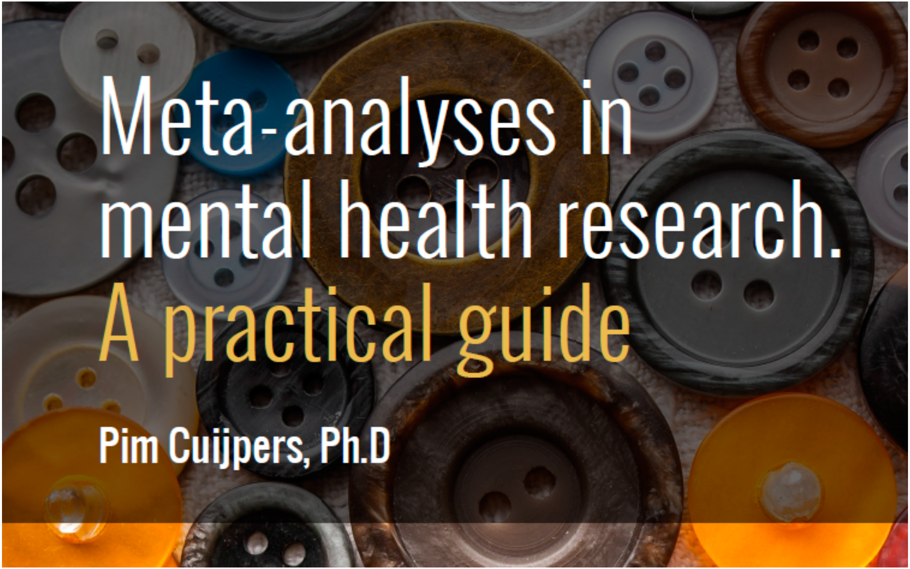
Thus from (34) and (35)

$$\hat{p} = \frac{\exp(\beta_0 + \beta_1 x_1 + \ldots + \beta_K x_K)}{1 + \exp(\beta_0 + \beta_1 x_1 + \ldots + \beta_K x_K)} \tag{36}$$

$$\ln\left(\frac{\hat{p}}{1-\hat{p}}\right) = \beta_0 + \beta_1 x_1 + \ldots + \beta_K x_K \tag{37}$$

Parameters interpretation in log-odds

Odds and risk ratios – online (link)



Odds and risk ratios – online (link)

Table 4.2 Possible dichotomous outcomes in a randomized controlled trial

	Event (success)	No event (fail)	Total
Therapy	a	b	(a+b)
Control	c	d	(c+d)

$$\textbf{Odds ratio (OR)} = \frac{(a^*d)}{(b^*c)} = \frac{\text{Odds of success in the treatment group}}{\text{Odds of success in the comparison group}}$$

$$\textbf{Relative risk (RR)} = \frac{a/(a+b)}{c/(c+d)} = \frac{\text{Risk of success in treatment group}}{\text{Risk of success in comparison group}}$$

Odds and risk ratios – online (link)

The **Odds Ratio** (OR) is somewhat more complicated to understand. The OR represents the "odds" that an event will occur in the treatment, compared to the "odds" of the event occurring in the comparison group. The "odds" itself is the ratio of the probability that a particular event will occur to the probability that it will not occur. The OR can be any number between zero and infinity.

The OR was developed as a statistic that resembled the RR. And just as is the case as with the RR, an OR of 1 indicates that odds for an event in the treatment group are the same as the odds in the comparison group and that when the 95% confidence interval of the OR does not include 1, the OR is significant at the $p<0.05$ level. The OR and RR are different concepts, but the value of the OR approximates the value of the RR when the event occurs in 10% of the cases or less.

Odds and risk ratios – online (link)

The **Relative Risk** (RR) is the "risk" of participants in the treatment group to have the event (or success), divided by the "risk" of participants in the comparison group to have the event. The "risk" is simply the proportion of participants that have the event. If the RR is 1, the risk for the event in the treatment and comparison group are exactly the same, so there is no difference between treatment and comparison group. If the 95% confidence interval of the RR does not include 1, then the RR is significant at the $p<0.05$ level.

Specification test in regression models

1. Ramsey's RESET test
   - $H_0: \qquad y_i = x_i\beta + \varepsilon_i,$
   - $H_0: \qquad y_i = f\left(x_i\beta\right) + \varepsilon_i,$
   - where $f(.)$ is a nonlinear function.
2. Step 1
   - Regression $y_i = x_i\beta + \varepsilon_i,$
   - Fitted values $\hat{y}_i$.
3. Step 2
   - Auxiliary regression $y_i = x_i\beta + \alpha_1\hat{y}_i^2 + \ldots + \alpha_p\hat{y}_i^{p+1} + u_i,$
   - Test $\alpha_1 = \ldots = \alpha_p = 0.$

LINKTEST

1. LINKTEST test
   - $H_0: \quad y_i = f(x_i\beta) + \varepsilon_i,$
   - $H_1: \quad y_i = g(x_i\beta) + \varepsilon_i,$
   - where $g(.) \neq f(.)$.
2. Step 1
   - Regression $y_i = f(x_i\beta) + \varepsilon_i,$
   - Fitted values $\hat{y}_i$.
3. Step 2
   - Auxiliary regression $y_i = \alpha_0 + \alpha_1\hat{y}_i + \alpha_2\hat{y}_i^2 + u_i,$
   - Test $\alpha_2 = 0$.

Logit/Probit - LINKTEST

```
# Linktest
# Step 1
mylogit <- glm(winner~nominations+gglobes, data=oscar,
               family=binomial(link="logit"))
oscar$yhat = log(mylogit$fitted.values/(1-mylogit$fitted.values))
# Step 2
oscar$yhat2 = oscar$yhat^2
aux.reg = glm(winner~yhat+yhat2, data=oscar, family=binomial(link="logit"))
summary(aux.reg)

Call:
glm(formula = winner ~ yhat + yhat2, family = binomial(link = "logit"),
    data = oscar)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.03406    0.44867  -0.076 0.939490
yhat         1.03227    0.28856   3.577 0.000347 ***
yhat2        0.01598    0.08737   0.183 0.854863
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Another example for Linktest

```
# Linktest
# Step 1
mylogit <- glm(winner~gglobes, data=oscar, family=binomial(link="logit"))
oscar$yhat = log(mylogit$fitted.values/(1-mylogit$fitted.values))
# Step 2
oscar$yhat2 = oscar$yhat^2
aux.reg = glm(winner~yhat+yhat2, data=oscar, family=binomial(link="logit"))
summary(aux.reg)

Call:
glm(formula = winner ~ yhat + yhat2, family = binomial(link = "logit"),
    data = oscar)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   0.3436     0.4039   0.851   0.3949
yhat          0.5097     0.3267   1.560   0.1188
yhat2        -0.2837     0.1699  -1.670   0.0949 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

linktest.R

```
source("linktest.R")

linktest_result = linktest(model)
summary(linktest_result)

> summary(linktest_result)
Call:
glm(formula = y ~ yhat + yhat2,
family = binomial(link = model$family$link))
Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.03406    0.44867  -0.076 0.939490
yhat         1.03227    0.28856   3.577 0.000347 ***
yhat2        0.01598    0.08737   0.183 0.854863
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Specification tests for binary choice models - [2] p. 787

1. If x_2 is omitted from a model containing x_1 and x_2, (i.e. $\beta_2 \neq 0$) then

$$plim\hat{\beta}_1 = c_1\beta_1 + c_2\beta_2, \tag{38}$$

   where c_1 and c_2 are complicated functions of the unknown parameters. The implication is that even if the omitted variable is uncorrelated with the included one, the coefficient on the included variable will be inconsistent.

2. If the disturbances in the underlying regression are heteroscdastic, then the maximum likelihood method estimators are inconsistent and the covariance matrix is inappropriate.

How to choose between logit and probit?

The main conclusions emerging from the Chen-Tsurumi paper are as follows, and they aren't all that encouraging:

- If the binary data that are being modelled are "balanced" (i.e., there is roughly a 50-50 split between the zero and one values), then none of the above information criteria are very effective at discriminating properly between the Logit and Probit models.
- If the data are "unbalanced", then only the DIC and AIC criteria are effective.
- The more information that is available about the higher moments of the underlying distribution of the binary data, the more effective are these criteria in the "unbalanced" case.
- Sample sizes of at least 1,000 or more are needed to be able to discriminate between the Logit and Probit models using this approach.
- If these information criteria don't help us very much, is there some other way to choose between the Logit and Probit specifications?

Source: http://davegiles.blogspot.com/ (link)

cut-off probability in binary choice models

- Maximising only sensitivity is trivial: Take an extreme threshold such that all subjects are considered positive. Then of course specificity is poor. So you take both, sensitivity and specificity simultaneously.
- One criterion is the Youden index: The sum of sensitivity and specificity has to be maximal. However, you can choose a different risk function weighting the trade-off between sensitivity and specificity according to your practical insight into the field of application.
- For example, if you want to diagnose a severe disease, you would prefer higher sensitivity over high specificity. But if the treatment for this disease has also side effects, you would not want to expose so many patients to the treatment by tolerating low specificity.

Source: https://stats.stackexchange.com/ (link)

Information Criteria Unveiled

- Akaike, H., 1973. Information theory and an extension of the maximum likelihood principle. In B. N. Petrov and F. Csaki (eds.), 2nd. International Symposium on Information Theory. Akademia Kiado, Budapest, 267-281.

$$AIC = -2\ ln(L) + 2k \tag{39}$$

- Akaike, H., 1978. A Bayesian analysis of the minimum AIC procedure. Annals of the Institute of Statistical Mathematics, 30, Part A, 9-14.
- Schwarz, G., 1978. Estimating the dimension of a model. Annals of Statistics, 6, 461-464.

$$BIC = SBC = -2\ ln(L) + k\ ln(N) \tag{40}$$

Source: `https://davegiles.blogspot.com/2013/07/information-criteria-unveiled.html`

Information Criteria Unveiled

- Some authors (packages) scale both of the terms in the AIC formula, by dividing each of them by the sample size, n. With this scaling, the name AICk is sometimes used.
- Obviously, it won't matter whether you use AIC or AICk, as long as:
  i. you adopt a fixed convention when comparing different values of the information criterion;
  ii. you compute them using the same sample of data; and
  iii. you use them only for ranking competing models, and not as cardinal measures.

Source: `https://davegiles.blogspot.com/2013/07/information-criteria-unveiled.html`

Information Criteria Unveiled

- Most of you will have used, or at least encountered, various "information criteria" when estimating a regression model, an ARIMA model, or a VAR model. These criteria provide us with a way of comparing alternative model specifications, and selecting between them.
- They're not test statistics.
- Rather, they're minus twice the maximized value of the underlying log-likelihood function, adjusted by a "penalty factor" that depends on the number of parameters being estimated. The more parameters, the more complicated is the model, and the greater the penalty factor.
- For a given level of "fit", a more parsimonious model is rewarded more than a more complex model. Changing the exact form of the penalty factor gives rise to a different information criterion.

Source: `https://davegiles.blogspot.com/2013/07/information-criteria-unveiled.html`

Maximum Likelihood Method

Let $f(x;\theta)$ denote the probability mass function for a discrete distribution with associated parameter θ.
Suppose $X_1, X_2, \ldots, X_n$ are a random sample from this distribution and $x_1, x_2, \ldots, x_n$ are the actual observed values.

Then, the *likelihood function* of θ is

$$\begin{aligned} L(\theta|x_1, x_2, \ldots, x_n) &= P(X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n) \\ &= P(X_1 = x_1)P(X_2 = x_2)\cdots P(X_n = x_n) \\ &= f(x_1;\theta)f(x_2;\theta)\cdots f(x_n;\theta) \\ &= \prod_{i=1}^{n} f(x_i;\theta) \end{aligned}$$

The likelihood function is a function of θ and is sometimes written as $L(\theta) = L(\theta|x_1, x_2, \ldots, x_n)$.

Source: *Mathematical Statistics with Resampling and R, str. 137.*

Maximum Likelihood Method

Let $f(x;\theta)$ denote the pdf of a continuous random variable with associated parameter θ.
Suppose $X_1, X_2, \ldots, X_n$ are a random sample from this distribution and $x_1, x_2, \ldots, x_n$ are actual observed values.

Then, the *likelihood function* of θ is

$$L(\theta|x_1, x_2, \ldots, x_n) = f(x_1;\theta)f(x_2;\theta)\cdots f(x_n;\theta)$$

A maximum likelihood estimate is a statistic $\hat{\theta}$ that maximises $L(\theta)$

$$L(\hat{\theta}) \geq L(\theta) \text{ for all } \theta. \tag{41}$$

Source: *Mathematical Statistics with Resampling and R, str. 139.*

MLE Example

- Suppose you buy a weighted coin in a magic shop. It looks a typical coin with heads and tails, but the coin is not necessarily fair. You flip the coin eight times and observe the sequence HHHTHTHT. What would be a good estimate for p, the probability of heads, based on this sequence?
- Let $X_i \sim Bern(p)$, where $X_i = 1$ indicates heads. We will assume that the flips are independent.

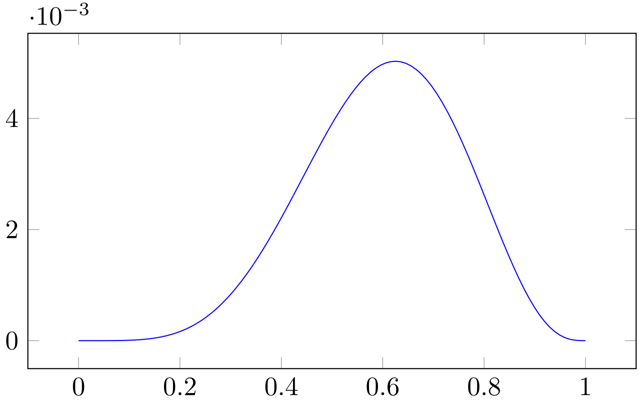
$$
\begin{aligned}
& P(X_1 = 1, X_2 = 1, X_3 = 1, X_4 = 0, X_5 = 1, X_6 = 0, X_7 = 1, X_8 = 0) \\
= \; & P(X_1 = 1) P(X_2 = 1) P(X_3 = 1) \cdots P(X_8 = 0) \\
= \; & p^5 (1-p)^3
\end{aligned}
$$

- $\hat{p}_{MLE} = 5/8$

Source: *Mathematical Statistics with Resampling and R, str. 136.*

MLE Example

$$
\begin{aligned}
& P(X_1 = 1, X_2 = 1, X_3 = 1, X_4 = 0, X_5 = 1, X_6 = 0, X_7 = 1, X_8 = 0) \\
= \quad & \ldots = p^5(1-p)^3
\end{aligned}
$$

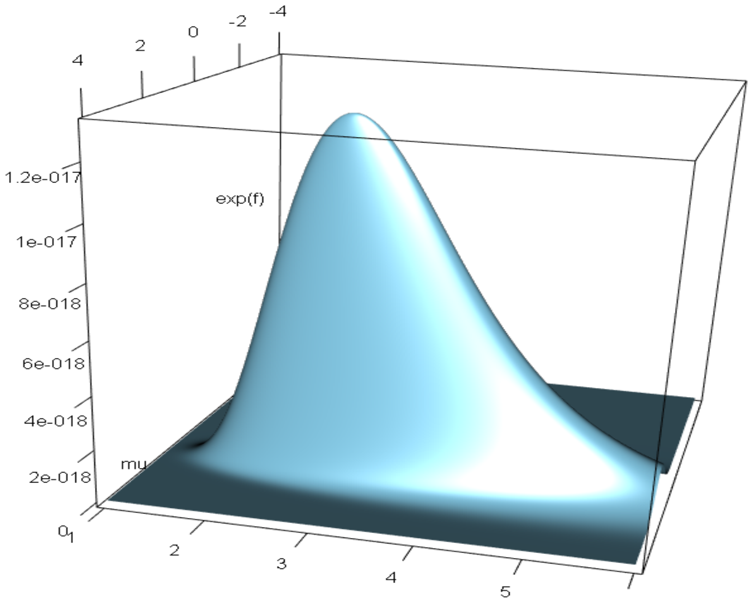


MLE Example II

Suppose that the observations
4.76, 0.35, 0.04, -1.26, 3.30, 3.79, 0.82, -1.18, -0.77, 2.47, 1.50, 2.62, 1.62, 3.27, 1.89, 1.45, 1.61, 2.78, -0.98, 2.41.
are i.i.d. sample from $N(\mu, \sigma^2)$ The likelihood function for $\theta = (\mu, \sigma)'$ is

$$\begin{aligned}
L(\mu, \sigma | x_1, \ldots, x_n) &= \prod_{i=1}^{20} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-(x_i - \mu)^2}{2\sigma^2}\right) \\
&= \left(\frac{1}{\sqrt{2\pi}}\right)^{20} \left(\frac{1}{\sigma}\right)^{20} \exp\left(\frac{-\sum_{i=1}^{20}(x_i - \mu)^2}{2\sigma^2}\right) \\
\ln L(\mu, \sigma | x_1, \ldots, x_n) &= l(\mu, \sigma | x_1, \ldots, x_n) = \\
&= -\frac{20}{2} \ln(2\pi) - 20 \ln(\sigma) - \frac{\sum_{i=1}^{20}(x_i - \mu)^2}{2\sigma^2}
\end{aligned}$$

MLE Example II



MLE Example II

```
install.packages("rgl")
library("rgl")

x = c(4.76,0.35,0.04,-1.26,3.30,3.79,0.82,-1.18,-0.77,2.47,
      1.50,2.62,1.62,3.27,1.89,1.45, 1.61,2.78,-0.98,2.41)
n = length(x)

loglik = function(mu, sigma2) {
  ll = -n/2*log(2*pi)-n/2*log(sigma2)-sum((x-mu)^2)/(2*sigma2)
  return(ll) }

mu = seq(from=-4, to=4, by=0.01)
sigma2 = seq(from=1, to=6, by=0.01)
f = matrix(0, nrow=length(mu), ncol=length(sigma2))

for(i in 1:length(mu)){
  for(j in 1:length(sigma2)){
    f[i,j] = loglik(mu[i],sigma2[j])
  }
}

persp3d(mu,sigma2,f)
image(mu, sigma2, f)
image(mu, sigma2, exp(f))
```

Maximising likelihood function

- Analytically
- Numerically
  1. Newton's method (Newton-Raphson)
  2. Method of Scoring
  3. BHHH Algorithm
  4. BFGS Algorithm
  5. Simulated Annealing
  6. Other methods

Likelihood function for the Oscar probit model

Let's assume a simple model for Oscar movies

$$oscar_i = \beta_0 + \beta_1 nominations_i + \varepsilon_i \tag{42}$$

Probability of the success is then

$$Prob(oscar_i = y_i) = \left[\Phi(\beta_0 + \beta_1 nom_i)\right]^{y_i} \left[1 - \Phi(\beta_0 + \beta_1 nom_i)\right]^{1-y_i} \tag{43}$$

Likelihood function is then

$$L(\beta_0, \beta_1) = \prod_{i=1}^{N} \left[\Phi(\beta_0 + \beta_1 nom_i)\right]^{y_i} \left[1 - \Phi(\beta_0 + \beta_1 nom_i)\right]^{1-y_i} \tag{44}$$

And the log-likelihood function

$$lnL(\beta_0, \beta_1) = \sum_{i=1}^{N} y_i \Phi(\beta_0 + \beta_1 nom_i) + \sum_{i=1}^{N} (1 - y_i) \left[1 - \Phi(\beta_0 + \beta_1 nom_i)\right] \tag{45}$$

Likelihood function for the Oscar probit model

Likelihood function is then

$$L(\beta_0, \beta_1) = \prod_{i=1}^{N} \left[\Phi(\beta_0 + \beta_1 nom_i)\right]^{y_i} \left[1 - \Phi(\beta_0 + \beta_1 nom_i)\right]^{1-y_i} \tag{46}$$

And the log-likelihood function

$$lnL(\beta_0, \beta_1) = \sum_{i=1}^{N} y_i \Phi(\beta_0 + \beta_1 nom_i) + \sum_{i=1}^{N} (1 - y_i) \left[1 - \Phi(\beta_0 + \beta_1 nom_i)\right] \tag{47}$$

Now, we have to maximise this function with regard to β_0 and β_1.

More on the topic

- To extend the material discussed I highly recommend these:
- https://statisticalhorizons.com/hosmer-lemeshow
- https://statisticalhorizons.com/alternatives-to-the-hosmer-lemeshow-test
- https://davegiles.blogspot.com/2012/06/another-gripe-about-linear-probability.html
- https://davegiles.blogspot.com/2011/05/gripe-of-day.html
- https://davegiles.blogspot.com/2013/05/robust-standard-errors-for-nonlinear.html#more

Literature

[1] R. Carter Hill, William E. Griffiths and Guay C. Lim, **Principles of Econometrics**, John Wiley & Sons.

[2] W.H. Greene, *Econometric Analysis*, 6th ed., Pearson Prentice Hall, 2008.